

EXPERIMENT-13

13. AIM: Study about Wireshark

Wireshark is an open-source packet analyzer, which is used for **education, analysis, software development, communication protocol development, and network troubleshooting.**

It is used to track the packets so that each one is filtered to meet our specific needs. It is commonly called as a **sniffer, network protocol analyzer, and network analyzer.** It is also used by network security engineers to examine security problems.

Wireshark is a free to use application which is used to apprehend the data back and forth. It is often called as a free packet sniffer computer application. It puts the network card into an unselective mode, i.e., to accept all the packets which it receives.

A packet is a unit of data which is transmitted over a network between the origin and the destination. Network packets are small, i.e., maximum **1.5 Kilobytes for Ethernet packets and 64 Kilobytes for IP packets.** The data packets in the Wireshark can be viewed online and can be analyzed offline

Installation of Wireshark Software

Below are the steps to install the Wireshark software on the computer:

- Open the web browser.
- Search for '**Download Wireshark.**'
- Select the Windows installer according to your system configuration, either 32-bit or 64-bit. Save the program and close the browser.
- Now, open the software, and follow the install instruction by accepting the license.
- The Wireshark is ready for use.

On the network and Internet settings option, we can check the interface connected to our computer.

If you are Linux users, then you will find Wireshark in its package repositories.

i. Packet Capture Using Wire shark

Capturing your traffic with Wireshark

After starting Wireshark, do the following:

1. Select **Capture | Interfaces**
2. Select the interface on which packets need to be captured. This will usually be the interface where the Packet/s column is constantly changing, which would indicate the presence of live traffic). If you have multiple network interface cards (i.e. LAN card and Wi-Fi adapter) you may need to check with your IT administrator to determine the right interface.
3. Click the **Start** button to start the capture.
4. Recreate the problem. The capture dialog should show the number of packets increasing. Try to avoid running any other internet applications while capturing, closing other browsers, Instant messengers etc.
5. Once the problem which is to be analyzed has been reproduced, click on **Stop**. It may take a few seconds for Wireshark to display the packets captured.
6. Save the packet trace in the default format. Click on the **File** menu option and select **Save As**. By default Wireshark will save the packet trace in libpcap format. This is a filename with a.pcap extension.

ii. Starting Wire shark using command line

You can start Wireshark from the command line, but it can also be started from most Window managers as well. In this section we will look at starting it from the command line.

Wireshark supports a large number of command line parameters. To see what they are, simply enter the command *wireshark -h*

Wireshark 3.7.3 (v3.7.3rc0-25-g72703582d587)
Interactively dump and analyze network traffic.
See <https://www.wireshark.org> for more information.

Usage: wireshark [options] ... [<infile>]

Capture interface:

-i <interface>, --interface <interface>
 name or idx of interface (def: first non-loopback)
-f <capture filter> packet filter in libpcap filter syntax

iii. Viewing Captured Traffic

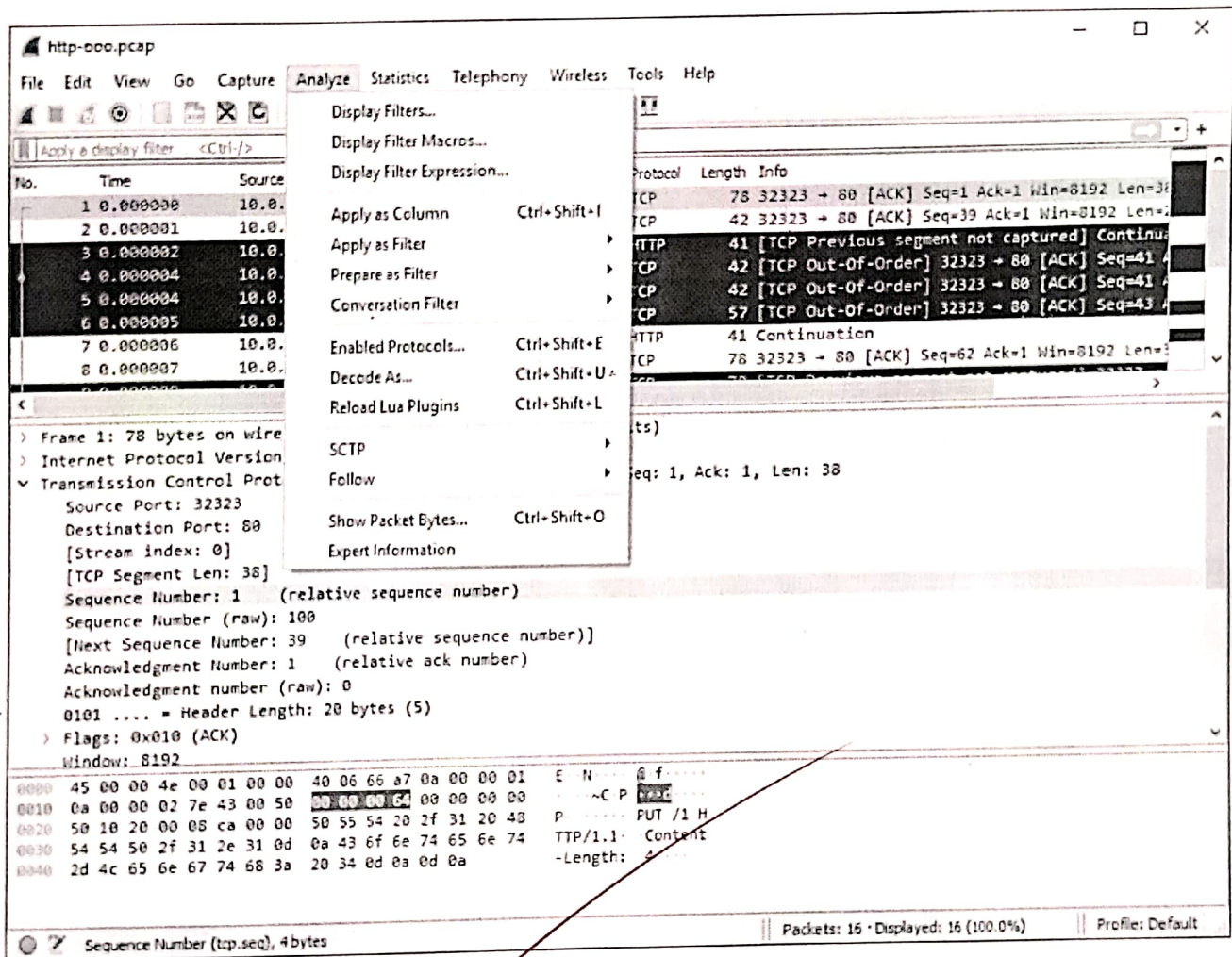
Once you have captured some packets or you have opened a previously saved capture file, you can view the packets that are displayed in the packet list pane by simply **clicking on a packet in the packet list pane**, which will bring up the selected packet in the tree view and byte view panes. Once you have captured some packets or you have opened a previously

saved capture file, you can view the packets that are displayed in the packet list pane by simply clicking on a packet in the packet list pane, which will bring up the selected packet in the tree view and byte view panes.

iv. Analysis and Statistics & Filters.

Analysis menu in wireshark

The “Analyze” Menu The Wireshark Analyze menu contains the fields shown in Analyze menu items.



The “Statistics” Menu in wireshark

The Wireshark Statistics menu contains the fields shown in [Table 3.9, “Statistics menu items”](#).

The screenshot shows the Wireshark interface with the following components:

- Menu Bar:** File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, Help.
- Toolbar:** Standard icons for file operations, capture, and analysis.
- Filter Bar:** Apply a display filter (Ctrl+F).
- Packet List:**

No.	Time	Source
1	0.000000	200.121.1.131
2	0.000011	172.16.0.122
3	0.025738	200.121.1.131
4	0.025749	172.16.0.122
5	0.076967	200.121.1.131
6	0.076978	172.16.0.122
7	0.102939	200.121.1.131
8	0.102946	172.16.0.122
9	0.120205	200.121.1.131
10	0.128319	172.16.0.122
11	0.154162	200.121.1.131
12	0.154169	172.16.0.122
13	0.179906	200.121.1.131
14	0.179915	172.16.0.122
- Packet Details:**
 - Frame 1: 1454 bytes on wire (11632 bits)
 - Ethernet II, Src: VMware_c0:00:01
 - Internet Protocol Version 4, Src: 200.121.1.131
 - Transmission Control Protocol, Src Port: 80 (80), Seq: 1, Ack: 1, Len: 1400
- Statistics Panel:**
 - Capture File Properties
 - Resolved Addresses
 - Protocol Hierarchy
 - Conversations
 - Endpoints
 - Packet Lengths
 - I/O Graph
 - Service Response Time
 - DHCP (BOOTP) Statistics
 - ONC-RPC Programs
 - 29West
 - ANCP
 - BACnet
 - Collectd
 - DNS
 - Flow Graph
 - HART-IP
 - HPFEEDS
 - HTTP
 - HTTP2
 - Sametime
 - TCP Stream Graphs
 - UDP Multicast Streams
 - IPv4 Statistics
 - IPv6 Statistics
- Packet Bytes:**

Offset	Hex	ASCII
0000	00 0c 29 42 12 13 00 50 56 c0 00 01 08 00 45 00	...)B...P V.....E.
0010	05 a0 01 41 00 00 6a 06 d3 90 c8 79 01 83 ac 10	...A...j. ...y....
0020	00 7a 29 3a 00 50 a7 5c 04 40 e2 e2 ee bf 50 10	..z):.P...H....P.
0030	ff ff 77 67 00 00 30 54 73 57 77 51 74 45 79 4e	...w...T slwQtEyh
0040	45 61 33 78 70 74 44 63 51 4f 2f 6b 75 31 41 52	E53xptDc Q0/kulAR
0050	52 66 47 59 67 53 32 41 34 47 59 35 31 56 33 32	RfGVgS2A 4GV51V32
- Status Bar:** Packets: 3083 - Displayed: 3083 (100%)

Filters in wireshark

Wireshark has two filtering languages: **capture filters** and **display filters**. Capture filters are used for filtering when capturing packets, "Filtering while capturing". Display filters are used for filtering which packets are displayed.



Filter: tcp

Expression... Clear Apply

No. -	Time	Source	Destination	Protocol	Info
11	1.226156	192.168.0.2	192.168.0.1	TCP	3196 > http [SYN] Seq=0 Len=0 MSS
12	1.227282	192.168.0.1	192.168.0.2	TCP	http > 3196 [SYN, ACK] Seq=0 Ack=
13	1.227325	192.168.0.2	192.168.0.1	TCP	3196 > http [ACK] Seq=1 Ack=1 Win
14	1.227451	192.168.0.2	192.168.0.1	HTTP	SUBSCRIBE /upnp/service/Layer3For
15	1.229309	192.168.0.1	192.168.0.2	TCP	http > 3196 [ACK] Seq=1 Ack=256 W
16	1.231421	192.168.0.1	192.168.0.2	TCP	http > 3196 [ACK] Seq=1 Ack=256 W
17	1.248355	192.168.0.1	192.168.0.2	TCP	1025 > 5000 [SYN] Seq=0 Len=0 MSS
18	1.248391	192.168.0.2	192.168.0.1	TCP	5000 > 1025 [SYN, ACK] Seq=0 Ack=
19	1.250171	192.168.0.1	192.168.0.2	HTTP	HTTP/1.0 200 OK
20	1.250285	192.168.0.2	192.168.0.1	TCP	http > http [FIN, ACK] Seq=256 Ac
21	1.250810	192.168.0.1	192.168.0.2	TCP	http > 3196 [FIN, ACK] Seq=114 Ac
22	1.250842	192.168.0.2	192.168.0.1	TCP	3196 > http [ACK] Seq=257 Ack=115
23	1.251868	192.168.0.1	192.168.0.2	TCP	1025 > 5000 [ACK] Seq=1 Ack=1 Win
24	1.252826	192.168.0.1	192.168.0.2	TCP	http > 3196 [FIN, ACK] Seq=26611
25	1.253323	192.168.0.2	192.168.0.1	TCP	http > http [SYN] Seq=0 Len=0 MSS
26	1.254502	192.168.0.1	192.168.0.2	TCP	http > 3197 [SYN, ACK] Seq=0 Ack=
27	1.254532	192.168.0.2	192.168.0.1	TCP	3197 > http [ACK] Seq=1 Ack=1 Win

+ Frame 11 (62 bytes on wire, 62 bytes captured)

+ Ethernet II, Src: 192.168.0.2 (00:0b:5d:20:cd:02), Dst: Netgear_2d:75:9a (00:09:5b:2d:75:9a)

+ Internet Protocol, Src: 192.168.0.2 (192.168.0.2), Dst: 192.168.0.1 (192.168.0.1)

+ Transmission Control Protocol, Src Port: 3196 (3196), Dst Port: http (80), Seq: 0, Len: 0

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0000  00 09 5b 2d 75 9a 00 0b 5d 20 cd 02 08 00 45 00  ..[-u... ] ....E.
0010  00 30 18 48 40 00 80 06 61 2c c0 a8 00 02 c0 a8  .0.H@....a;....?.
0020  00 01 0c 7c 00 50 3c 36 95 f8 00 00 00 00 70 02  ...|.P<6.....p.
0030  fa f0 27 e0 00 00 02 04 05 b4 01 01 04 02      .....

```